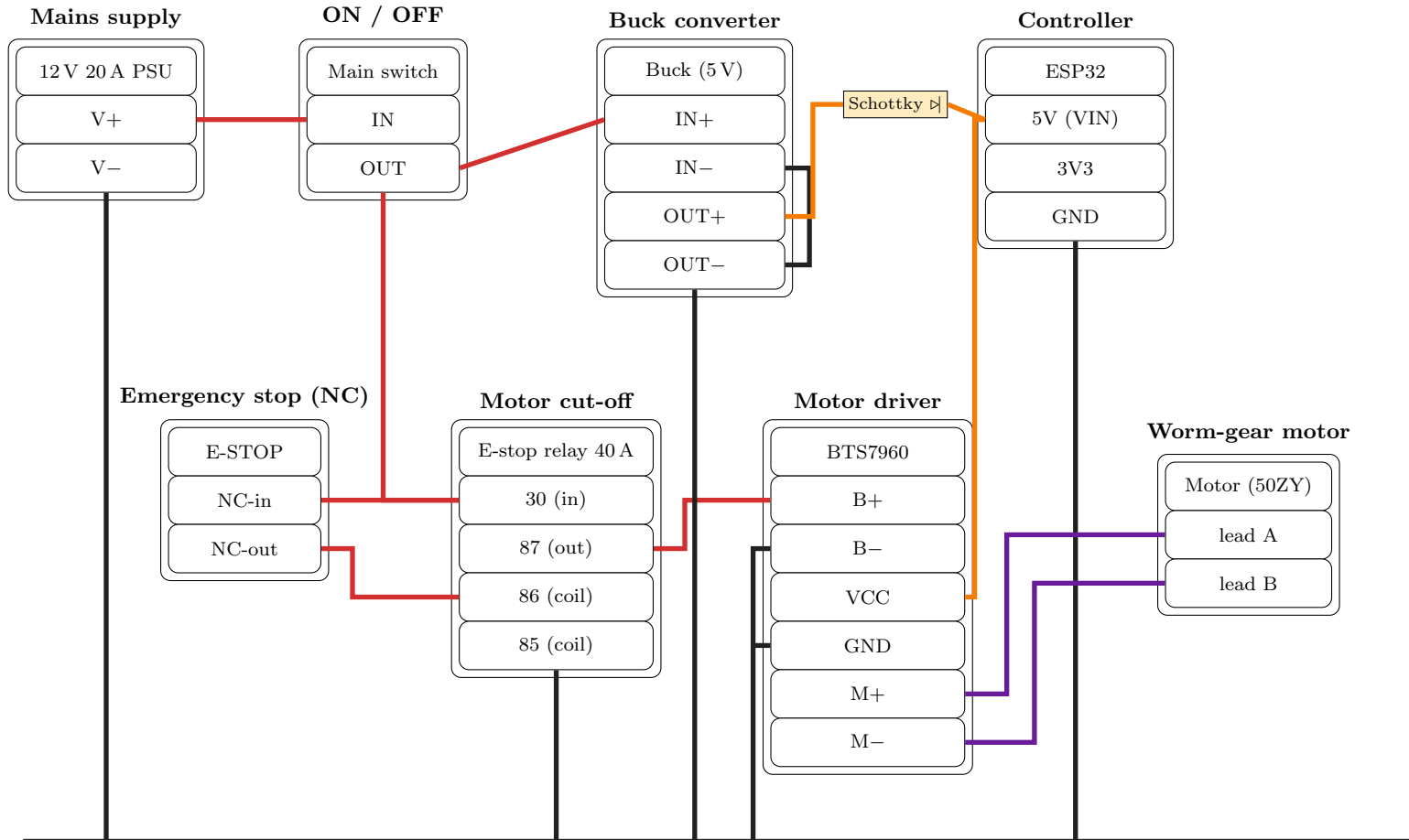


SolarMax — Wiring Diagrams

Visual companion to `wiring.md` (the tables are the authoritative per-pin detail). Wires colored by net; all grounds are one common ground. Pin names match `include/config.h`.

Wire color legend: — +12 V — GND — +5 V — +3.3 V — Motor out — Motor PWM/EN — I²C — Limit switch — Anemometer sig — Analog (pot/LDR)

1. Power Distribution



How the safety chain works

Main switch breaks 12 V+ to everything (ON/OFF).

E-stop (red mushroom, NC) sits in the relay *coil*: pressing it — or any broken wire — drops the coil, the relay opens, and **motor power is cut**. Fail-safe. The self-locking worm gear then holds the panel.

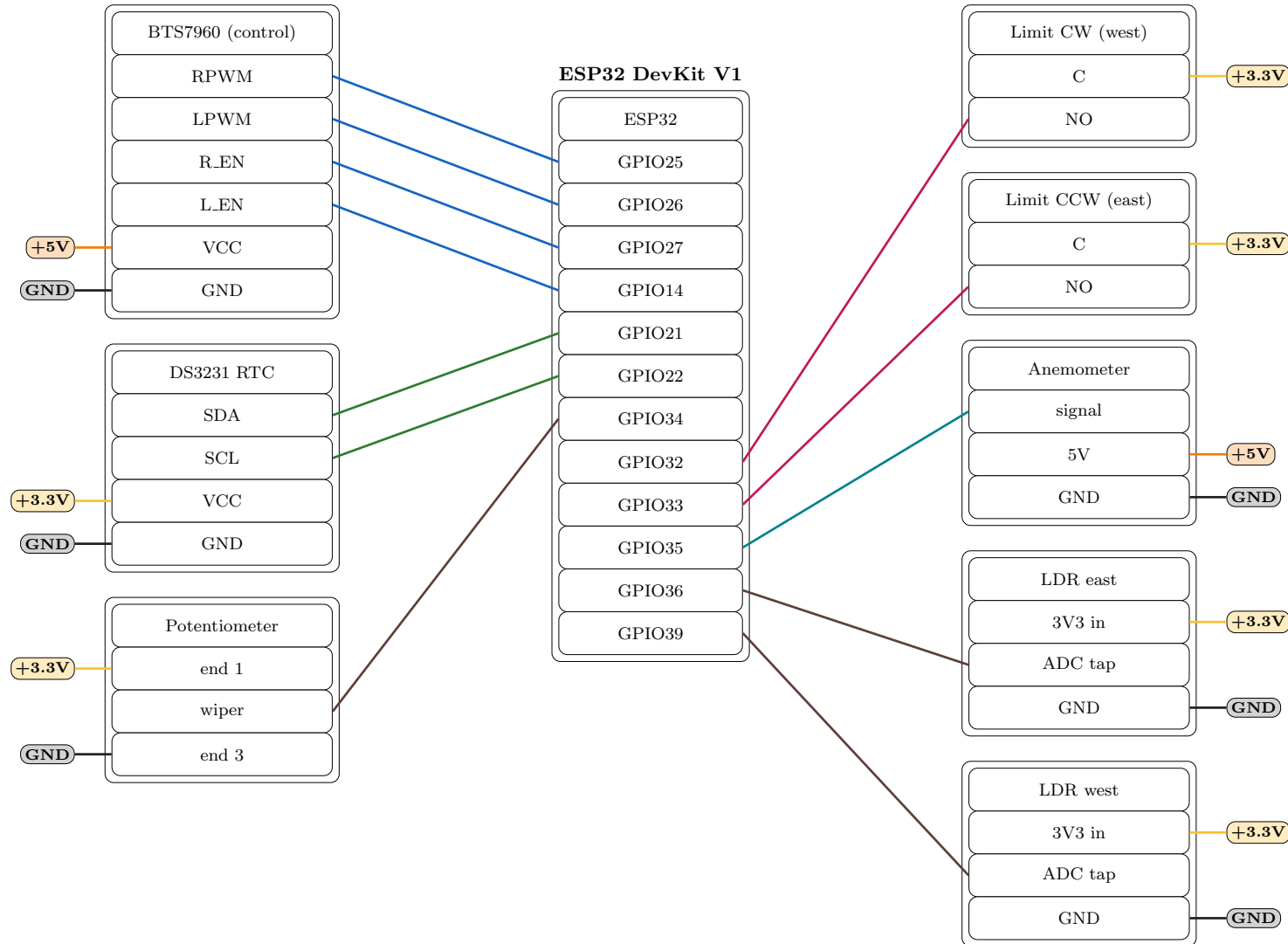
The E-stop button carries only the small coil current; the **40 A relay** carries the ~19 A motor current. Use **12 AWG** on the motor path.

Powering / programming

The ESP32 runs from the buck via its **5V pin**. The **Schottky diode** lets you plug USB in any-time to flash without conflict. **USB = programming only**.

2. Control & Sensor Wiring

ESP32 in the centre, mirroring the DevKit's two pin headers: **motor + feedback on the left**, **sensors on the right**. Power/ground pins tag out to the far edges. **Every GND tag is the one common ground.**

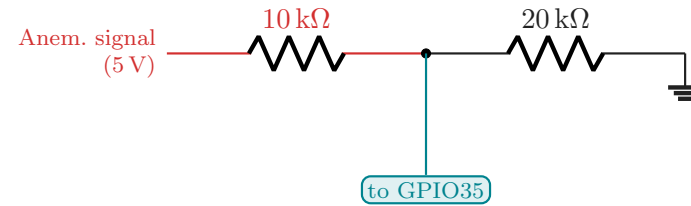


Every **GND** tag is the one common ground. **+5V** and **+3.3V** come from the rails in Diagram 1. Limit switches: C → +3.3V, NO → GPIO (no GND pin). The LDR and anemometer signal taps are the centre of the resistor dividers detailed in Diagram 3.

3. Real Circuits — Dividers & 5 V Rail Diode

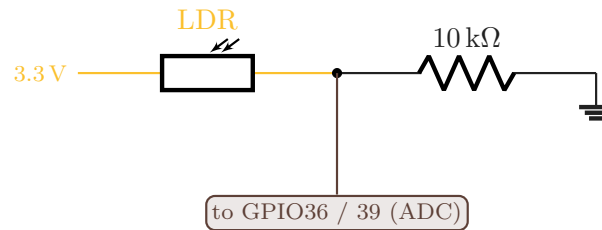
Drawn with proper component symbols (this is the part that becomes a KiCad schematic). Build these exactly — they protect the ESP32 and set correct readings.

(a) Anemometer signal: 5 V $\rightarrow$ 3.3 V



$$V_{out} = 5\text{ V} \times \frac{20}{10 + 20} \approx 3.3\text{ V}$$

(b) LDR (each of two): 3.3 V — LDR — ADC — 10 kΩ — GND



More light $\rightarrow$ lower LDR resistance $\rightarrow$ higher ADC reading

(c) 5 V rail diode (lets USB stay pluggable)



stripe (cathode) toward the ESP — blocks backfeed to USB